

Investigation Report:

`fleet_sensitivity_dual.png` and Related Figure Issues

Summary

`fleet_sensitivity_dual.png` is **NOT broken**. It exists at the correct path (`results/figures/fleet_sensitivity_dual.png`), contains valid dual-panel chart content (Mean Response Time vs Fleet Size + Coverage vs Fleet Size), and is correctly referenced in all documentation files.

The **actual broken figure** was `capacity_sensitivity_heatmap.png` , which had a **policy label mapping bug** causing P0 data to be silently dropped. This has been fixed.

Step 1: References to `fleet_sensitivity_dual`

Found in **5 documentation files**:

File	Reference Type
<code>docs/technical_report.md</code> (line 472)	<code>![Fleet Sensitivity Analysis](/home/ubuntu/ems-optimization/results/figures/fleet_sensitivity_dual.png)</code>
<code>docs/technical_report.md</code> (line 947)	Listed as Figure 3 in artifact index
<code>docs/visualization_index.md</code> (line 105)	Catalog entry with source script <code>run_production_v2.py</code>
<code>docs/figure_trace_guide.md</code> (line 91)	Trace entry #30 linking to technical report Section 5.6
<code>docs/output_comparison_report.md</code> (line 124)	Listed under fleet sensitivity figures

All references use the **correct path**: `results/figures/fleet_sensitivity_dual.png` .

Step 2: Actual Files Found

Three fleet sensitivity files exist:

File	Size	Content
<code>fleet_sensitivity_dual.png</code>	285 KB	✓ Valid dual-panel chart (2777×1231 px)
<code>fleet_sensitivity_curve.png</code>	65 KB	Valid single-panel curve
<code>fleet_sensitivity_v2_dual.png</code>	(variant)	Valid chart (not referenced in docs)

No mismatch found — the referenced file exists and renders correctly.

Step 3: Root Cause of the Reported Issue

The actual problematic figure was `capacity_sensitivity_heatmap.png`, which was documented in `output_comparison_report.md` as:

`capacity_sensitivity_heatmap.png` — Capacity sensitivity heatmap (**broken — shows “Data format issue”**)

Bug Details

In `scripts/generate_capacity_sensitivity_heatmap.py`, the policy label mapping was:

```
POLICY_LABELS = {
    "P0_spatial": "P0",    # Expected 'P0_spatial' in data
    "P1_demand": "P1",
    "P2_optimised": "P2",
}
```

But the simulation CSV (`results/capacity_comparison/simulation_results.csv`) stores the policy as `"P0"`, not `"P0_spatial"`. This caused:

1. P0 rows failed the `map()` → returned `NaN`
2. `dropna(subset=["Policy"])` silently removed all P0 data
3. Heatmap rendered with only P1 and P2 (or showed “Data format issue” in earlier versions)

Fix Applied

Updated `POLICY_LABELS` to handle all known variants:

```
POLICY_LABELS = {
    "P0": "P0",
    "P0_spatial": "P0",
    "P1": "P1",
    "P1_demand": "P1",
    "P2": "P2",
    "P2_optimised": "P2",
}
```

Also fixed the `reindex` to deduplicate values:

```
policy_order = list(dict.fromkeys(v for v in POLICY_LABELS.values() if v in pivot.index))
```

Step 4: Verification

After the fix, the heatmap correctly shows all three policies:

K=20:

Policy	cap=1	cap=2	cap=3	cap=4	cap=5
-----	-----	-----	-----	-----	-----
P0	3.11	3.11	3.11	3.11	3.11
P1	2.59	2.62	2.62	2.62	2.62
P2	2.56	2.56	2.56	2.56	2.56

K=40:

Policy	cap=1	cap=2	cap=3	cap=4	cap=5
-----	-----	-----	-----	-----	-----
P0	2.44	2.44	2.44	2.44	2.44
P1	2.39	2.32	2.34	2.34	2.34
P2	2.38	2.42	2.46	2.50	2.49

Step 5: Other Referenced Files — Path Mismatch Audit

While investigating, I found **44 images referenced in docs** that exist in subdirectories of `results/` but not in `results/figures/`. These are not broken — the docs reference them with correct relative paths to their actual locations (e.g., `results/production_v2/figures/`, `results/capacity_comparison/`, etc.). No action needed for these.

No other scripts had the `P0_spatial` vs `P0` label mismatch.

Files Changed

File	Change
scripts/generate_capacity_sensitivity_heatmap.py	Fixed POLICY_LABELS mapping + deduplicated reindex
results/figures/capacity_sensitivity_heatmap.png	Regenerated with all 3 policies
docs/output_comparison_report.md	Removed “broken” annotations

Commit: fix: capacity_sensitivity_heatmap missing P0 due to policy label mismatch